

Project Design (Updated for CSE499B Final Submission):

AI Medical Pre-Screening Assistant

From a 5-Stage ASR Pipeline to a 15-Module Conversational Clinical Intelligence System

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1 Introduction

This chapter presents the **updated** design of the *AI Medical Pre-Screening Assistant*. The original CSE499A design was a five-stage one-shot pipeline: Speech → ASR → NER → Structured EHR → Doctor Dashboard. Based on empirical findings from Phases 2 and 3 and clinical NLP requirements, the design has been expanded to **15 functional modules** that add an adaptive conversational follow-up loop, emergency detection, risk classification, explainable AI, and a continuous feedback and retraining pipeline. The system remains strictly a *pre-screening and decision-support tool*; final clinical authority rests with the attending physician.

2 Design Comparison: Original vs. Updated

Dimension	Original (5 stages)	Updated (15 modules)
Interaction	One-shot; no follow-up	Conversational loop until profile is complete
Emergency	Absent	M5: rule-based + AI; immediate escalation
Info gaps	Absent	M6: checklist per symptom category
Risk level	Hints only	M10: Low/Med/High/Critical + rule engine
Explainability	Absent	M11 XAI: SHAP/LIME plain-language drivers
Report	Basic EHR	M12: 8-section report; PDF + dashboard
Feedback loop	Absent	M15: annotations retrain M3, M7, M10
Bangla support	ASR only	End-to-end bilingual Bangla/English

3 Updated System Architecture — 15 Modules

3.1 Complete Pipeline Diagram

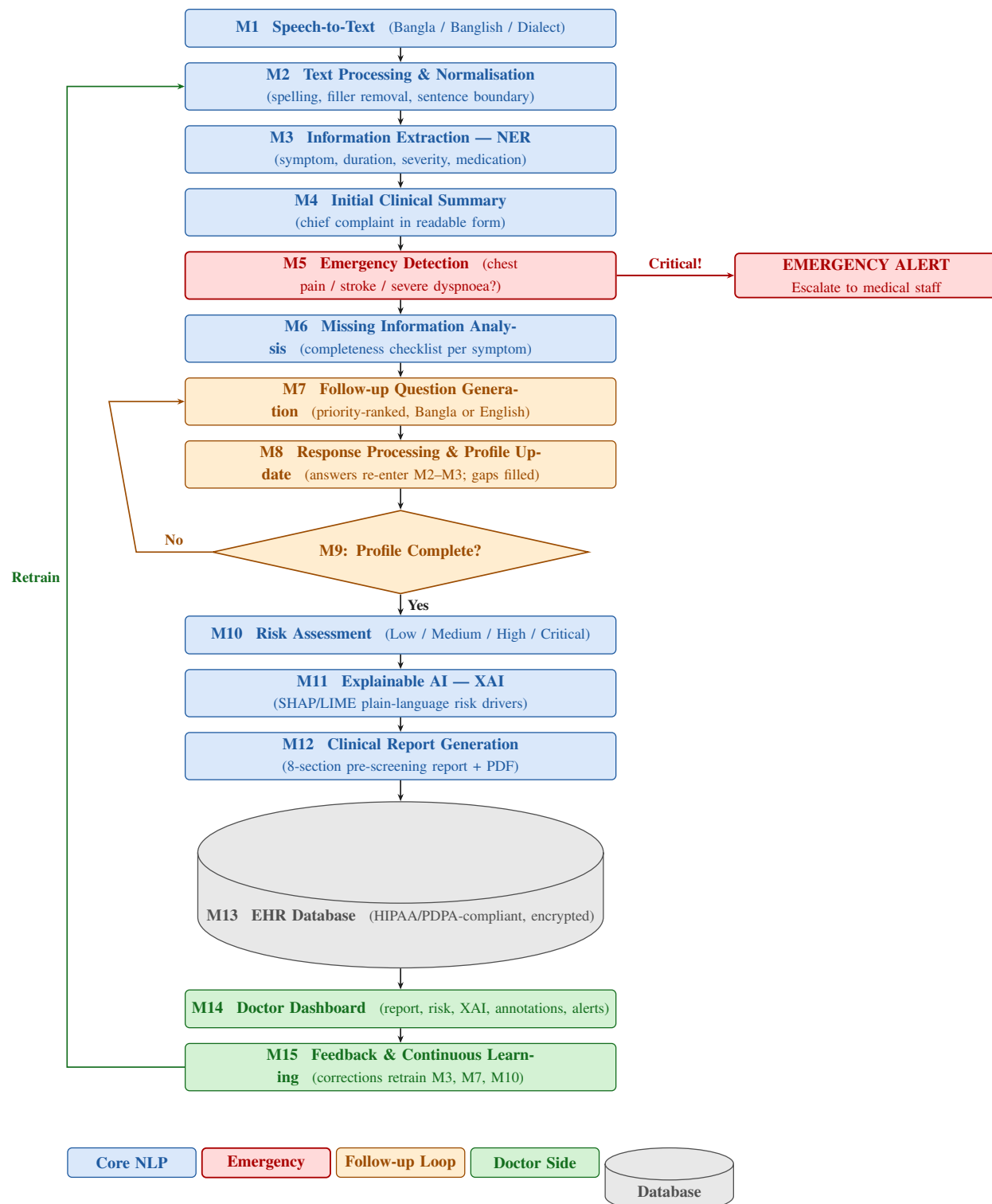


Figure 1: Complete 15-module pipeline (top-to-bottom). Orange loop (M7–M9) repeats until the patient profile is complete. Red path (M5) escalates emergencies immediately. Green arrow (M15) returns doctor feedback for model retraining.

3.2 Module Descriptions

M1 — Speech-to-Text.

Bangla/Banglish/dialect audio; VAD-segmented to 10–15 s clips. Best: *BengaliAI Regional* (46.94% WER);

CSE499B target <20% WER via LoRA on Qwen3-ASR-1.7B.

M2 — Text Processing & Normalisation.

Spelling correction, Banglish normalisation, filler removal, punctuation restoration, sentence-boundary detection.

M3 — Information Extraction (NER).

BanglaBERT labels: *symptom, site, duration, severity, frequency, age, comorbidity, medication*; temporal expressions normalised to standard intervals.

M4 — Initial Clinical Summary.

Human-readable chief-complaint summary, e.g. *“Fever + dry cough 3 days; mild fatigue; no chest pain.”* Shown on dashboard.

M5 — Emergency Detection (new).

Rule-based checker + AI classifier. Red flags: chest pain, stroke signs, severe dyspnoea, loss of consciousness, acute trauma. Escalates immediately; pipeline continues in parallel.

M6 — Missing Information Analysis (new).

Per-symptom clinical checklist: ☒ Fever detected; ☐ Duration missing; ☐ Temperature not provided.

M7 — Follow-up Question Generation (new).

Priority-ranked questions in Bangla or English. Example: *“Apnar jwor ki matro kal theke?”* / *“When did the fever start?”*

M8 — Response Processing & Profile Update (new).

Patient answers re-enter M2–M3; extracted entities merged into profile with conflict resolution; filled gaps marked.

M9 — Case Completion Check (new).

Scores completeness; if incomplete loops to M7; if complete (or max turns) proceeds to M10.

M10 — Risk Assessment (new).

Low / Medium / High / Critical via clinical rule engine weighted by age, comorbidities, severity, duration. Integrates M5.

M11 — Explainable AI (XAI) (new).

SHAP/LIME in plain language. Example: *“High risk: fever >5 days + dyspnoea + age >65.”*

M12 — Clinical Report Generation (new).

8-section report: Patient Profile, Chief Complaint, Symptoms, Emergency Flags, Q&A Log, Risk+XAI, Condition Categories, Next Steps.

M13 — EHR Database.

HIPAA/PDPA-compliant PostgreSQL; encryption at rest and in transit; full audit logging.

M14 — Doctor Dashboard (extended).

Report, risk, flags, XAI. Annotation/override of any field (versioned); real-time alerts for High/Critical.

M15 — Feedback & Continuous Learning (new).

Doctors rate reports, correct extractions, annotate risk disagreements; periodic retraining of M3, M7, M10 with regression testing.

4 System Flowchart — Patient Journey

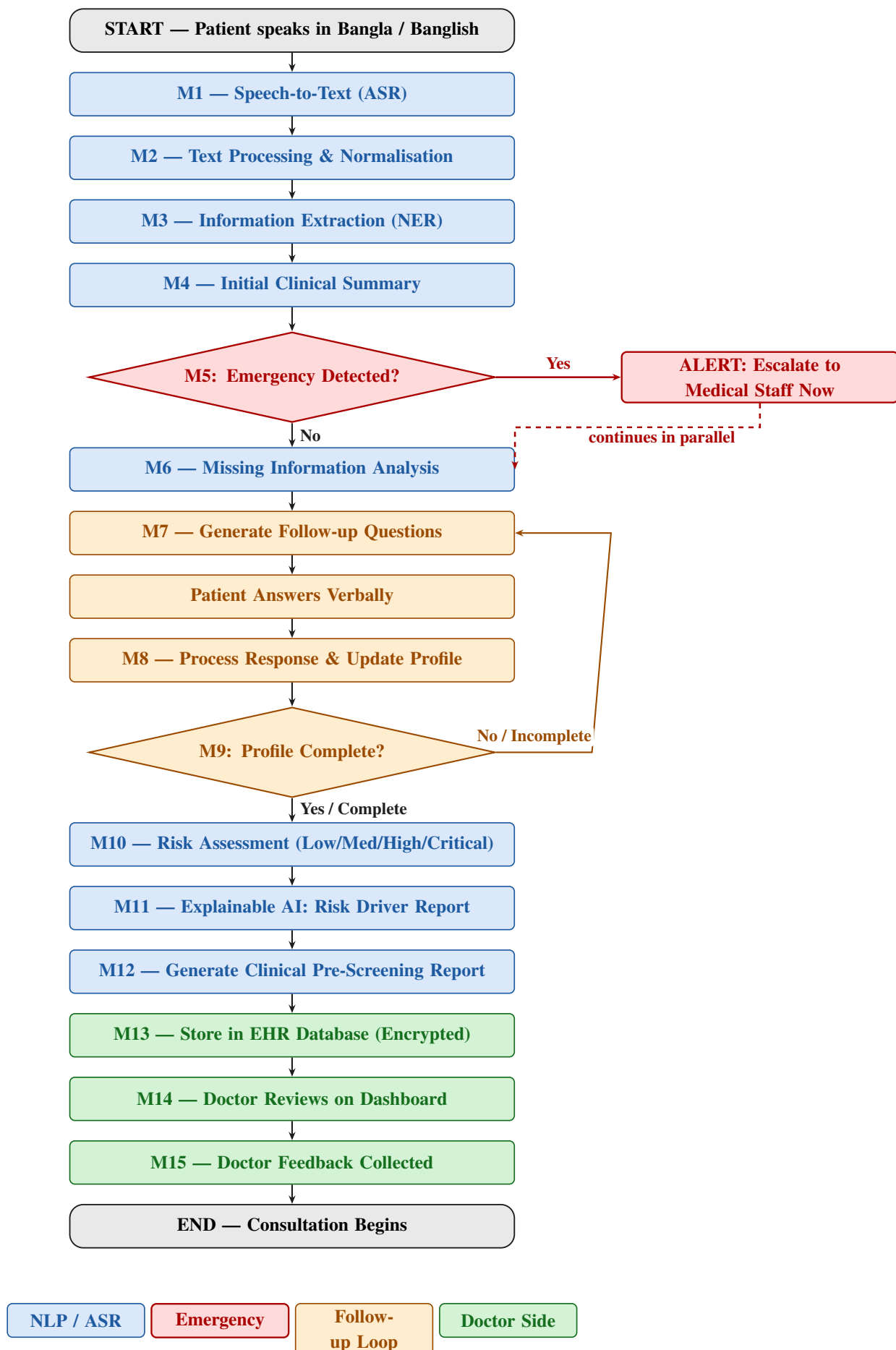


Figure 2: Patient journey flowchart across all 15 modules (top-to-bottom). The orange loop (M7–M9) iterates until the clinical profile is complete. The red path (M5) alerts staff immediately; the dashed line shows the pipeline continues in parallel.

5 Software Components

Tool	Function	Rationale
Python 3.10	Implementation	lan- Standard ML/NLP ecosystem
PyTorch 2.x	Deep-learning	frame- Supports all benchmarked ASR models
HF Transformers	Model loading & inference	Unified API: Whisper, Wav2Vec2, MMS, Qwen-Audio
HF PEFT/Accelerate	LoRA fine-tuning	Parameter-efficient tuning of Qwen3-ASR-1.7B (CSE499B)
BanglaBERT	Medical NER — M3	Best Bangla NLP accuracy
WhisperX	VAD segmentation — M1	Handles audio >30 s
LIME / SHAP	XAI attribution — M11	Standard explainability libraries
FastAPI	Backend REST/WebSocket	Async; low latency for follow-up loop
PostgreSQL	EHR store — M13	Encryption-at-rest; HIPAA-ready
React / Next.js	Doctor dashboard — M14	Responsive UI; role-based access
Flutter / PWA	Patient kiosk — M1	Single button; Bangla voice prompts
MLflow	Retraining pipeline — M15	Experiment tracking; model versioning
jiwer	WER/CER/BLEU evaluation	Standard ASR metrics
Google Colab (T4)	GPU compute	Free tier for inference benchmarks

6 CSE499A Implementation Phases

Phase 1 — Audio Collection & Preprocessing

Demo: 12 March 2026. Multi-dialect corpus (Dhakaiya, Barishali, Sylheti, Normal, Indian Bangla); ≈4.7 h, 10 sources, ~1,100 clips after WhisperX VAD segmentation at 16 kHz mono PCM.

Phase 2 — Baseline ASR Benchmark (12 models)

Demo: 2 April 2026. Best: **BengaliAI Regional** — 46.94% WER, 24.29% CER, 0.273 BLEU. Key finding: dialect speech defeats all open Bangla models.

Phase 3 — Large Multimodal LLM Benchmark

Demo: 9 April 2026. Qwen2-Audio, Qwen3-ASR-1.7B, Voxtral-Mini, Phi-4, Qwen2.5-Omni. None matched BengaliAI Regional (>50 pp WER gap). Conclusion: *specialisation beats scale for Bangla dialect ASR*.

Phase 4 — Fine-Tuning Methodology Study

Demo: 16 April 2026. LoRA adapter plan for Qwen3-ASR-1.7B: adapter on speech encoder, decoder frozen. Training scheduled for CSE499B.

Phase 5 — Comparative Chatbot Output Study

Execution: next demo. ChatGPT, DeepSeek, Qwen, Perplexity evaluated on simulated patient utterances; results inform Module 12 report schema.

7 Conclusion

The 15-module design addresses every limitation of the original five-stage pipeline: the adaptive follow-up loop (M6–M9) guarantees clinical information completeness; M5 adds an emergency escalation path;

M10–M11 add risk stratification and explainability; M15 closes the quality loop through physician-driven retraining. The CSE499B priority is LoRA fine-tuning of Qwen3-ASR-1.7B to reach <20% dialect WER; Modules 3–15 will then be integrated iteratively.

References

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